

Digital Signal Filtering - Hard(+) Problem

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This is a **very hard** problem. Only attempt this problem if you feel it will not be a significant burden for you to complete. Otherwise, I suggest picking something else.

Provided for your in the same folder where you found this pdf is a giant text file, which contains voltage vs time data. The time is in microseconds, and the voltage is in, well, volts. This is a signal, which contains data that you must decode. Here is some information about this data:

The carrier wave for the signal of interest is 500hz. There is binary data encoded using amplitude shift keying (a method of encoding binary data which can be googled). A "1" is encoded by 0-5V swing in the amplitude of the 500hz signal. A "0" is encoded by a 2-3V swing (basically less than a full 0-5). 20 periods of the carrier wave indicates a single bit. Unfortunately, it is not as simple as just decoding the signal this way though. There is also noise, specifically 100hz and 1000hz random amplitude signals added on top. There is also a bit of noise applied across the spectrum. The binary encoding is ASCII, with the MSB first. Also, for your convenience, the sample rate of the data is 20000hz. This can also be seen of course by just looking at the timestamps for each datapoint.

Your goal is to decode the ASCII message hidden within the signal. Write a program that takes this text file as input and provides the encoded text output. You will know it worked if you get the right message.